Lecture 1: Introduction to data structures and algorithms

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2026

Outline

- Data structures and algorithms
- Industrial problems
- Data types
- Abstract data structures
- Problem description

Computer programs

- **What is a computer program?**

  A computer program is a set of instructions that perform specific tasks when executed by computers.

- **What is a good computer program?**
  - Correctness
  - Efficiency
  - Simple/Understandable
  - Easy to change/maintain and upgrade

"Program = Algorithms + Data structures"

N. Wirth

Data structures?

- What is a data structure?

  A data structure is a particular way of storing and organizing data so that they can be used efficiently.

- What is a good data structure?
  - Correctness (satisfy requirements)
  - Memory efficiency
  - Simple/understandable
  - Implementable

- Standard data structures
  - Array
  - Set
  - Map/Dictionary

Algorithms?

- What is an algorithm?

  An algorithm is a step by step procedure to solve a problem. An algorithm can be described by nature languages (English, Vietnamese...) or programming languages (C++, Java, Python…)

  Example: How to cook rice?

  - Step 1: Get rice
  - Step 2: Rinse the rice
  - Step 3: Put the rice and water into a rice cooker
  - Step 4: Turn on the rice cooker
  - Step 5: Check the rice when the rice cooker turns off
- What is a good algorithm?
  - Correctness
  - Efficiency
  - Simple/understandable
  - Implementable

Example: Searching problem

Problem: Given a list of numbers sorted increasingly, describe an algorithm to check if a number *k* is in the list.

➢ *Naïve searching algorithm*

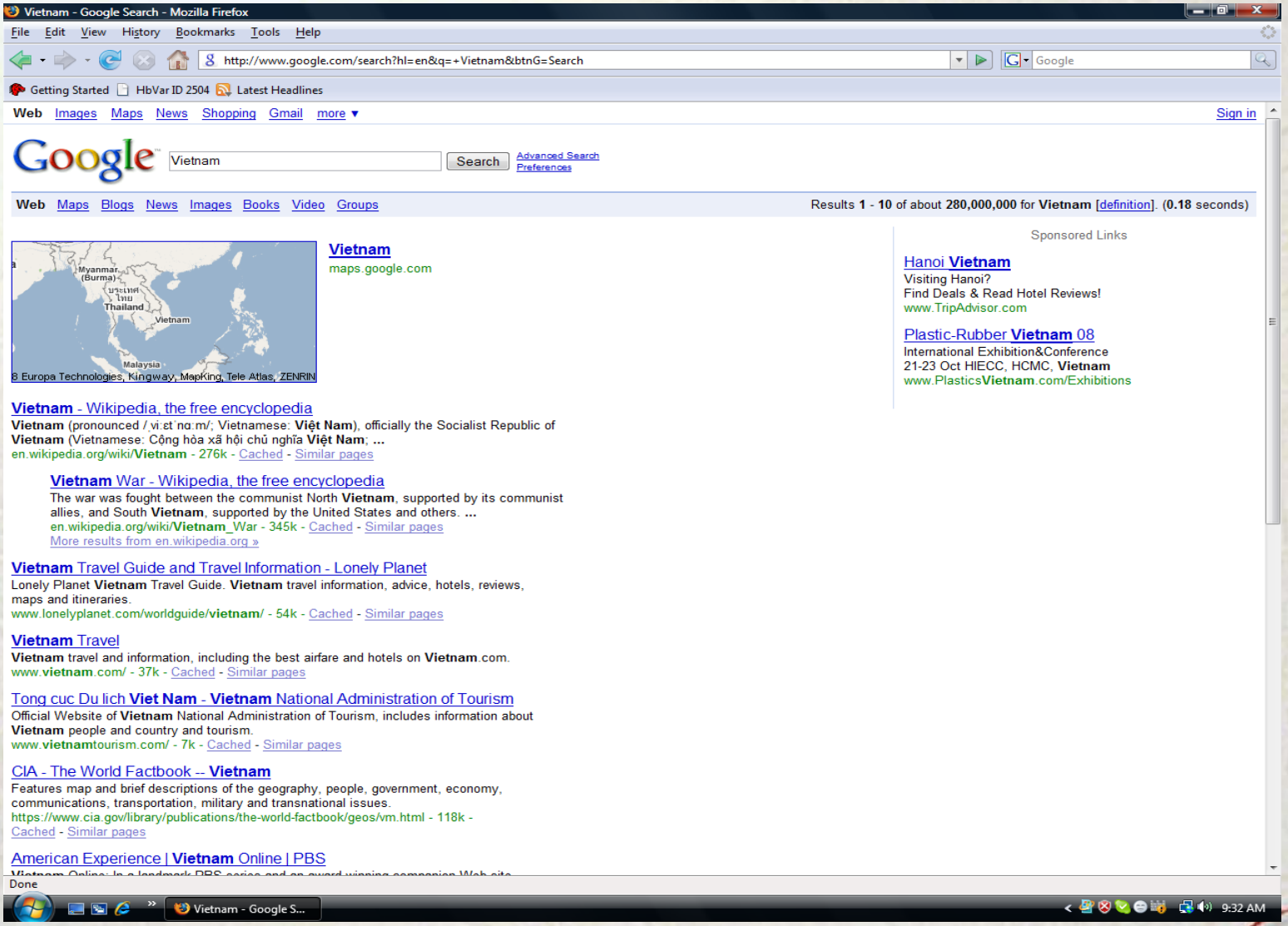
Moving from the begin to the end of the list to check if *k* is in the list.

➢ *Binary search algorithm*

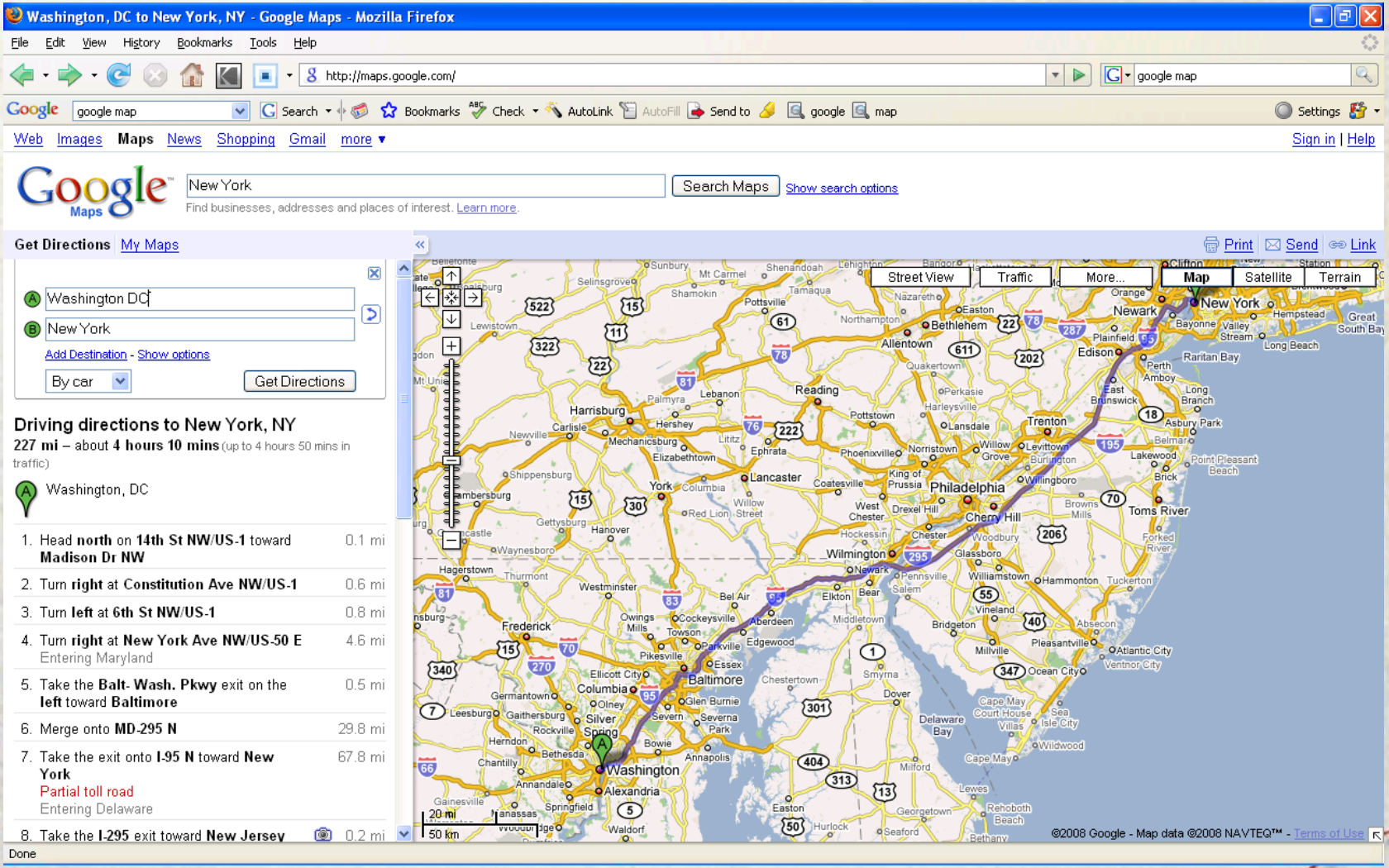
Compare *k* to the number at the middle of the list:

- ❖ if they are equal, finish the searching process.
- ❖ if *k* is smaller, perform the search in the first half of the list
- ❖ if *k* is greater, perform the search in the last half of the list

Google search



Find the shortest path (Google map)



Traveling salesman problem (TSP)

A salesman needs to visit *n* customers at *n* different locations. Find the shortest path for the salesman such as he visits each location exactly once and finally returns to the start location.



Quiz 1: The roles of data structures and algorithms





Quiz 2: The roles of data structures and algorithms





Data types

- A data type is specified by:
  - The range of values or the values it can hold.
  - Operations on values

type	range	operations
bool	true / false	and, or, not,
char	[-127, 127]	‘<‘, ‘>’, ‘=’, ‘+’, ‘-’, ‘*’, ‘/’
int	-32,767 -> 32,767	‘<‘, ‘>’, ‘=’, ‘+’, ‘-’, ‘*’, ‘/’
float	~1E-37 -> ~1E+37	‘<‘, ‘>’, ‘=’, ‘+’, ‘-’, ‘*’, ‘/’
double	~1.7E-308 -> ~1.7E+308	‘<‘, ‘>’, ‘=’, ‘+’, ‘-’, ‘*’, ‘/’

Structured data types

- Question: What data type to present a point on the plane?

- Solution: Programming languages provide rules to define a new data type T from defined data types

Example in C++:

```
struct  PersonType {
            string    name;
            int       age;
            bool      gender;
};
```

Structured data types

- How to present a point on the plane?

```
struct   PointType {
    double    x;
    double    y;
};
```

- How to present a line on the plane?

```
struct  LineType {
        PointType        start;
        PointType        end;
};
```

Structured data types

- Define a data structure to present student information

```
struct  StudentType {
  string   name;
  int      age;
  bool     gender;
}
```

- Define a data structure to present our class information

```
struct  ClassType{
   string          className;
   int             numberStudents;
   StudentType     students[100];
}
```

Range of structured data types

- The range of a structured data type T is determined by the ranges of data types that are used to construct T.

  Example:

```
struct  ComplexType {
            double    real;
            double    image;
};
```

- The range of ComplextType:
  - real: The range of ‘double’ type
  - image: The range of ‘double’ type

Operations on structured data types

- Operations on a structured data type of a program are defined by the program.

Example:

```
struct  ComplexType {
        double      real;
        double      image;
};
ComplexType  add (const ComplexType  &c1, const  ComplextType &c2) {
        ComplexType  sum;
        sum.real = c1.real + c2.real;
        sumimage = c1.image + c2.image;
        return sum;
}
ComplexType  multiply (const ComplexType  &c1, const  ComplextType &c2) {
        ComplexType  product;
        product.real = (c1.real * c2.real) – (c1.image * c2.image);
        product.image = (c1.real * c2.image) + (c1.image * c2.real);
        return product;
}
```

Abstract data types

- Abstract components of a structured data type

  Example: ComplexType
  - real //the real component of complex number
  - image //the image component of complex number

- Abstract operations of a structured data type

  Example: ComplexType
  - add (c1, c2): Add two complex numbers c1 and c2 and return their sum.
  - multiply (c1, c2): Multiply two complex numbers c1 and c2 and return their product.

Abstract data types

- Abstract Student data type
  - Abstract data type components
    - ID
    - Name
    - DOB
    - Address
  - Abstract operations
    - initializeStudent (ID, Name, DOB, Address)
    - compareStudents (student1, student2)
    - getID (student)
    - getName (student)
    - getDOB (student)
    - getAddress (student)

Quiz 3

- Abstract TEACHER data type
- Abstract COMPUTER data type
- Abstract RECTANGLE data type

Problem description

- **Description:**
  - What is the nature of problem?
  - What questions need to be solved?
  - What data and constrains are given?
  - What results are expected?
  - Examples
- **Input:** Describe the input and their specific constrains
- **Output:** Describe the output and their specific requirements

Birthday cake selection

- **Problem:** Today is Anna's birthday and she is going to buy a birthday cake. The cake shop has many birthday cakes with different sizes. As a kid, Anna wants to pick the largest one. For example, if there are 4 birthday cakes with sizes 1, 5, 3, and 7, Anna will pick the cake with size 7. Your task is to write a program to find the largest birthday cake for Anna.
- **Input:** Data come from file BirthdayCakes.txt as following:
  - The first line contains a single integer number *n,* denoting the number of birthday cakes at the shop ($0 < n < 10000$).
  - The second line contains *n* space-separated integer numbers each describes the size of one cake.
- **Output:** Write to file BirthdayCakes.out an integer number which is the size of the largest birthday cake.

Example:

BirthdayCakes.txt	BirthdayCakes.out
4 1 5 3 7	7

Quiz 4

- Describe the problem of finding the product of two big integer numbers
- Describe the problem of sorting students according to their final examinations
- Describe the problem of finding the shortest path between two locations